EENG 385 - Electronic Devices and Circuits
BJT Curve Tracer: 555 Timer
Lab Document

Outcome and Objectives

The outcome of this lab is to analyze, simulate and assemble a circuit that generates a periodic square wave using a 555 Timer and compare the expected behavior of the circuit from each mode of analysis. Through this process you will achieve the following learning objectives:

- Use a software tool to perform time and frequency domain analysis of an electronic circuit.
- Assemble a circuit on a PCB using the equipment in the laboratory.
- Use laboratory test and measurement equipment to analyze electronic circuits.

Analysis: 555 Timer

We will start by examining the behavior of today's circuit and then use our understanding of the circuit to quantify the behavior of the circuit. The circuit shown Figure 1 is built around the 555 Timer, a chip seen since the early 1970s. Its use is now limited to niche applications; it is not something you are likely to use much in industry. That said, an analysis of the circuit Figure 1 will allow us to practice valuable skills.

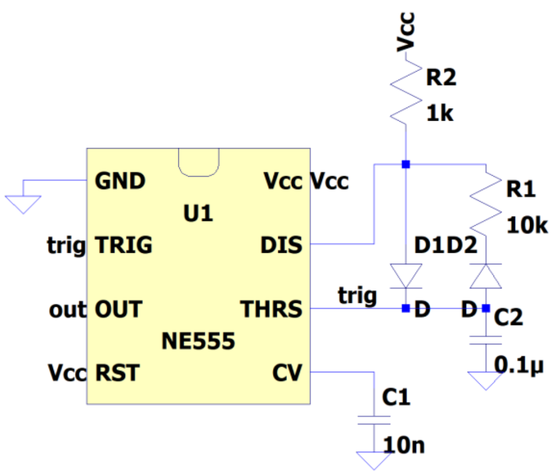


Figure 1: The output of the 555 Timer oscillates with a predictable period and duty cycle. Schematic drawn in LTspice.

In order to analyze the circuit in Figure 1, you first need to understand the internal operation of the 555 Timer chip. A highly simplified view of its internal structure is shown in Figure 2. You will notice some of the pins shown in Figure 1 are not shown in Figure 2. These omitted pins do not affect the behavior of the circuit shown in Figure 1, so to simplify the explanation, they are omitted.

Finally, note that VCC is the system voltage in Figure 1, for simplicity, we will assume throughout the lab that VCC = 9V.

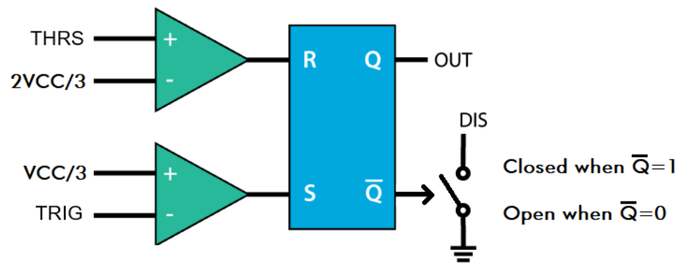


Figure 2: A highly simplified rendering of a 555 Timer's internals.

Let's start our exploration of Figure 2 by examining the pair of green triangles called comparators. We will call the upper comparator the "R-comparator" because it is connected to the R input of the blue box. Likewise, we will call the lower comparator the "S-comparator". Notice each comparator has a + and – input on its left, and its output on the right. The behavior of a comparator is described by the following two statements.

- When the voltage on the + input is larger than the voltage on the – input, the output of the comparator equals VCC.
- When the voltage on the - input is larger than the voltage on the + input, the output of the comparator equals GND.

Use these definitions and the schematic in Figure 2 to answer the following two questions. Assume that VCC = 9V, so state your answers in terms of numerical volts, not symbolically in terms of VCC.

1. For what range of voltages on THR does the R-comparator output VCC?
2. For what range of voltages on TRI does the S-comparator output VCC?

Notice the THR and TRI inputs to the 555 Timer are tied together in Figure 1. We will call this common voltage V_t in the following text. Make note of this assignment now. Next, let's look at the blue box in the middle of Figure 2 known as an SR-latch. The input voltages to the SR-latch come from the comparator. When the comparator outputs VCC, the SR-latch interprets this as a logic 1. A comparator output of GND is interpreted by the SR-latch as logic 0.

Table 1: State table for an SR-latch

S	R	Q	$\bar{Q}$	Name
0	0	Q	$\bar{Q}$	Hold
0	1	0	1	Reset
1	0	1	0	Set
1	1	X	X	Forbidden

The last element of Figure 2 is the voltage-controlled switch connected to the $\bar{Q}$ output of the SR-latch. This switch opens and closes under the control of the $\bar{Q}$ signal, hence the arrow head. When $\bar{Q} = 1$, the switch is closed, connecting the DIS pin to ground. When $\bar{Q} = 0$, the switch is open, leaving the DIS pin "floating", not connected to any voltage. In reality, this voltage-controlled switch is an NPN bipolar junction field effect transistor. However, since we have not studied them, I have replaced it with something more intuitive.

3. Now, let's summarize our understanding of the components in Figure 2 to complete Figure 3. In this figure, the vertical axis represents the voltage of V_t (the common voltage of the TRI and THR pins in Figure 1). The voltage range of V_t is split into 3 ranges, 9V-6V, 6V-3V, and 3V-0V. For each of these ranges:
    - Describe the logic values on the S and R inputs of the SR-latch. (You already did this in a previous question.) Write 0 or 1 as the value of S and R.
    - Use the S, R logic levels to determine the Q output. Write 0, 1, or Q as the value of Q.
    - Use the Q output to determine the state of the switch. The value of the switch should be "open", "closed," or "hold". Use "hold" when SR-latch is holding its output.
    - Use the switch value to determine the state of the DIS signal. The value of the DIS should be "open", "GND," or "hold". Use "hold" when the Switch is holding its output.

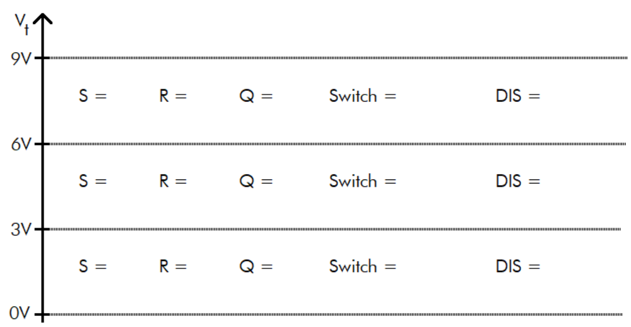


Figure 3: Determine the effect of V_t on the SR latch. Assume that VCC = 9V.

With our new understanding of the 555 Timer, let's return to the circuit shown in Figure 1 so that we can understand the role of the external components in the operation of the circuit. We will approach this question by replacing the 555 Timer with the V_t and DIS signals examined in Figure 3. The resulting circuit, shown in Figure 4, contains resistors, capacitors, a 555 Timer, and D1, D2, diodes. We will study diodes in much more detail later, but for now, consider them as circuit elements allowing current to flow in one direction as indicated by the triangle. So, for example, D1 will only allow current to flow downwards and D2 will only allow current to flow upwards.

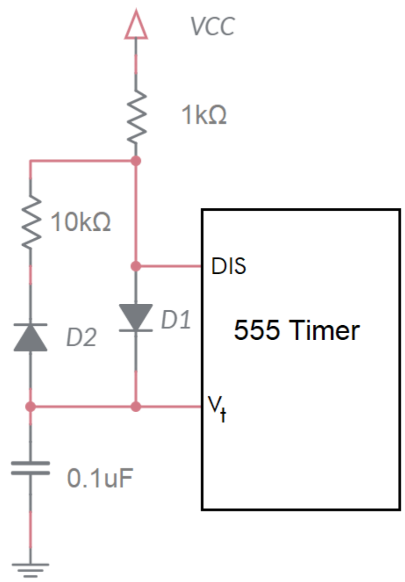


Figure 4: The circuit in Figure 1 with the 555 Timer replaced with the two signals examined in Figure 3.

Now, you will use the information from Figure 3 to determine to describe what the external circuit in Figure 4 functions like as V_t changes. Complete the sentence or write an expression for each of the bulleted items in questions 4 and 5. When two **bold** choices are separated by a "/", replace them with the correct answer. Fill in underlines with values.

4. Determine the circuit behavior in Figure 4 when V_t is between 0V-3V.
   - If V_t is between 0V-3V, then DIS is **open/closed** (Q=1).
   - If DIS is **open/closed,** then current wants to flow from VCC through D1 to ground.
   - This current flow will **charge/discharge** the capacitor with time constant RC = ______.
   - The equation describing the **charging/discharging** capacitor is $V_t(t) = 9(1 - e^{-t/\_\_\_\_})$. Fill in the denominator of the exponent.
   - Determine the time required for the capacitor to **charge** from 3V to 6V.
     - Set the $V_t(t)$ equation equal to 3V and solve for *t*. This result is the time to charge from 0V to 3V. Represent your answer in microseconds and round to the nearest integer.
     - Set the $V_t(t)$ equation equal to 6V and solve for t. This result is the time to charge from 0V to 6V. Represent your answer in microseconds and round to the nearest integer.
     - To get the time to **charge/discharge** from 3V to 6V, subtract the time to get to 3V from the time to get to 6V.

5. Determine the circuit behavior in Figure 4 when V_t is between 6V-9V.
   - If V_t is between 6V-9V, then DIS is **open/closed**, (Q=0).
   - If DIS is **open/closed,** then charge on the capacitor wants to flow through D2 to the grounded DIS.
   - This current flow will **charge/discharge** the capacitor with time constant RC = ________.
   - The equation describing the **charging/discharging** capacitor is $V_t(t) = 9e^{-t/\_\_\_\_}$ Fill in the denominator of the exponent.

- Determine the time required for the capacitor to **charge/discharge** from 6V to 3V.
  - Set the $V_t(t)$ equation equal to 6V and solve for t. This result is the time to discharge from 9V to 6V. Represent your answer in microseconds and round to the nearest integer.
  - Set the $V_t(t)$ equation equal to 3V and solve for t. This result is the time to discharge from 9V to 3V. Represent your answer in microseconds and round to the nearest integer.
  - To get the time to **discharge** from 6V to 3V, subtract the time to get to 6V from the time to get to 3V.

Let's now put all the pieces together to see how the external components and the internal structure of the 555 Timer, together, produce a periodic waveform. To do this, we will use Figure 5.

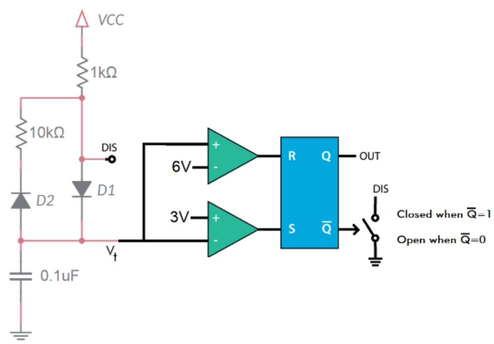


Figure 5: External circuits and internal organization of the 555 Timer .

You will examine the behavior of the system in terms of V_t. Complete the following set of statements using Figure 5 and the answers you provided to previous questions. When two answers are separated by a "/", circle the correct answer. Fill in underlines with values.

6. Assume V_t is less than 3V.
   a. Then S =____ and R =____ .
   b. Then Q = 1 and $\overline{Q} = 0$.
   c. Then DIS is **open/closed.**
   d. Then the capacitor is **charging/discharging.**
   e. Then it takes ____us to **charge** the capacitor from 3V to 6V.

So, whenever V_t drops below 3V, the capacitor starts charging, increasing V_t.

7. Assume V_t is greater than 6V.
   a. Then S =____ and R =____ .
   b. Then Q = 0 and $\bar{Q} = 1$.
   c. Then DIS is **open/closed.**
   d. Then the capacitor is **charging/discharging.**
   e. Then it takes ____µs to **discharge** the capacitor from 6V to 3V.

So, whenever V_t rises above 6V, the capacitor starts discharging, decreasing V_t.

8. Assume V_t is between 3V and 6V. The answers are given to you so that you can better follow what the circuit does.
    a. Then S = 0 and R = 0.
    b. Then Q and $\bar{Q}$ are unchanged.
    c. Then DIS remains unchanged.
    d. Then the capacitor charging state remains unchanged until V_t goes below 3V or above 6V

So, whenever V_t is between 3V and 6V, the capacitor continues behaving as it was before V_t entered the range 3V to 6V.

Note, V_t continuously oscillates between charging and discharging. As soon as V_t increases above 6V, the SR-latch changes state causing V_t to decrease. As soon as V_t drops below 3V, the SR-latch changes state causing V_t to start increasing. The cycle continues indefinitely.

9. Plot V_t vs. *time*. To help you, I have plotted voltage vs. time for V_t for the charging and discharging states. You can cut-and-paste the relevant pieces of the curves from Figure 6 to form your answer. Draw as many charge/discharge cycles as will fit.

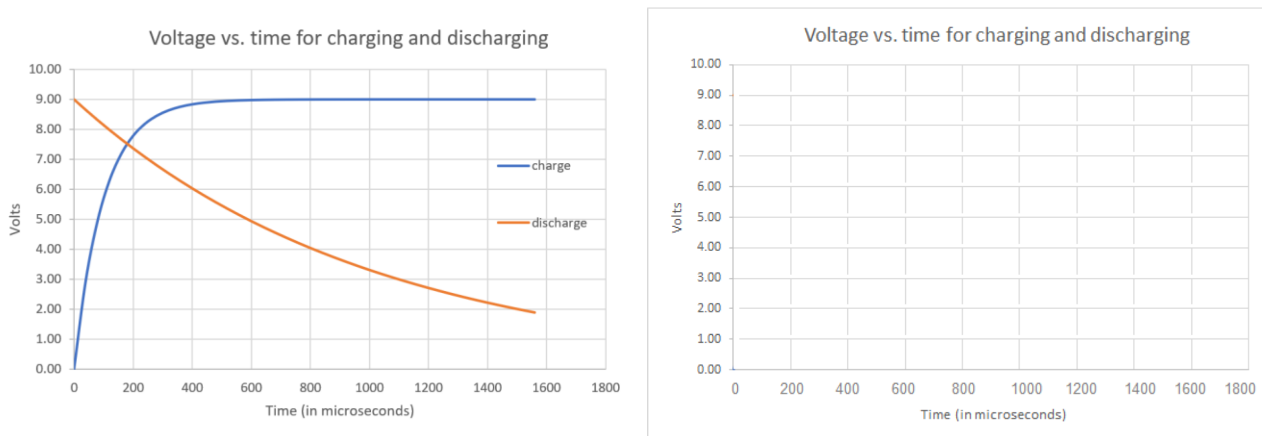


Figure 6: (Left) Voltage vs. time for V_t in both the charging and discharging configurations. (Right) Plot the charge/discharge curve of the capacitor using the information in the left graph.

10. Use this information to determine the rise time, fall time, and period of the waveform. Note the rise time corresponds to the output of the 555 Timer being high. The fall time corresponds to the output of the 555 Timer being 0V.
11. Use the period to determine the frequency of the waveform.
12. The duty cycle of a waveform is the percentage of time when it is at a high voltage level. What is the duty cycle of the OUT pin?

Simulation: 555 Timer

Use the skills learned in Lab 0 to build the 555 Timer circuit analyzed in the previous section to build the circuit shown in Figure 7 in LTspice.

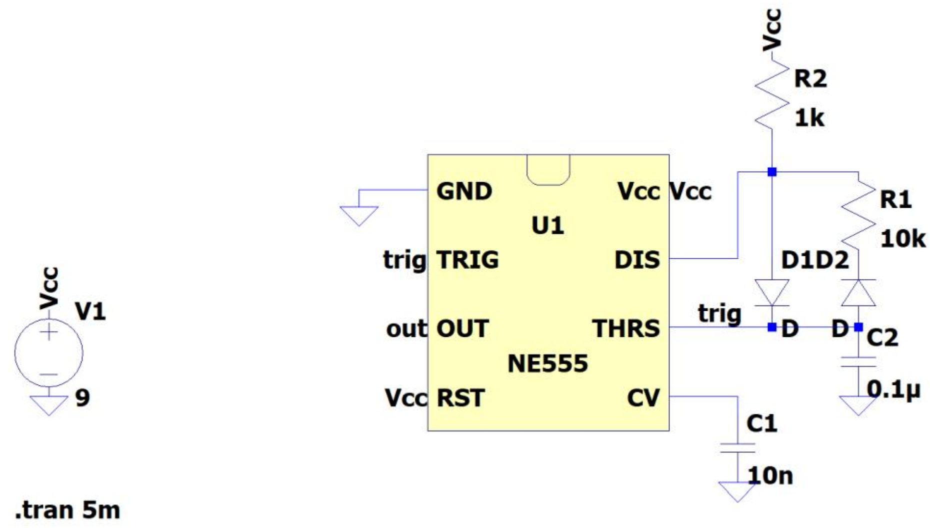


Figure 7: 555 Timer circuit to construct in LTspice.

You will find the components to build the LTspice simulation using the information listed in the table below.

- Make sure to give the resistors, capacitors and voltage supply the correct values.
- You can find the NE555 by searching for it in the component pop-up
- Use the components at the top of the LTspice window
- Make sure to use the Label Net tool to label the TRIG, OUT, RST, THRS and Vcc outputs

1. Simulate your circuit for 5 ms and while probing OUT and TRIG
    a. Click the green play icon at the top of the window
    b. In the Configure Analysis pop-up
        i. Select the Transient tab
        ii. Enter “5m” in the Stop time text box
        iii. Click OK
        You should see an empty waveform appear on the screen
    c. Add probes to your schematic
        i. Select the schematic window
        ii. Move the cursor to the OUT pin – it will transform into a red probe icon
        iii. While the cursor looks like a probe icon left mouse click
        You should see the OUT wave appear in the waveform window
        iv. Add a probe to the TRIG net
2. Capture the schematic and waveforms for your report
    a. Select the schematic
    b. Click on the Zoom to Fit icon at the top of the window
    c. Tools -> Copy bitmap to Clipboard
    d. Do the same for the TRIG waveform
3. Use the timing diagram to measure the time high, time low, period, frequency, and duty cycle of the waveform on the OUT pin.
    a. Select the waveform window

   b. Left click on "V(out)" at the top of the waveform window
      You should see white cursors appear in the waveform window along with a pop-up containing numerical data.
   c. Move the vertical cursor to the positive edge of the OUT wave
   d. Right mouse click in the waveform window and select Place Cursor on Active Trace. A second white cursor should appear along with additional data in the pop-up
   e. Position the second cursor on the negative edge just after the cursor you just placed
   f. Record the values into the time high of the waveform into Table 4.
   g. Manipulate the cursor to get the other values in Table 4.
4. Use the timing diagram to measure the high and low voltages of the TRIG pin.
   a. Remove the V(out) cursors by right clicking in the waveform window and selecting Clear All Cursors
   b. Click on V(trig) at the top of the waveform window.
   c. Repeat cursor steps used to measure OUT waveform and record your values in Table 4.

Empirical: 555 Timer

This week, you will be soldering in the components associated with the POWER INPUT, 555 TIMER subsystems shown in the Figure 8 schematic. It is important you do not solder in any of the other components in any of the other subsystems. This lab document contains cursory coverage of the assembly process. You can find much more detail in the Assembly Guide posted on Canvas.

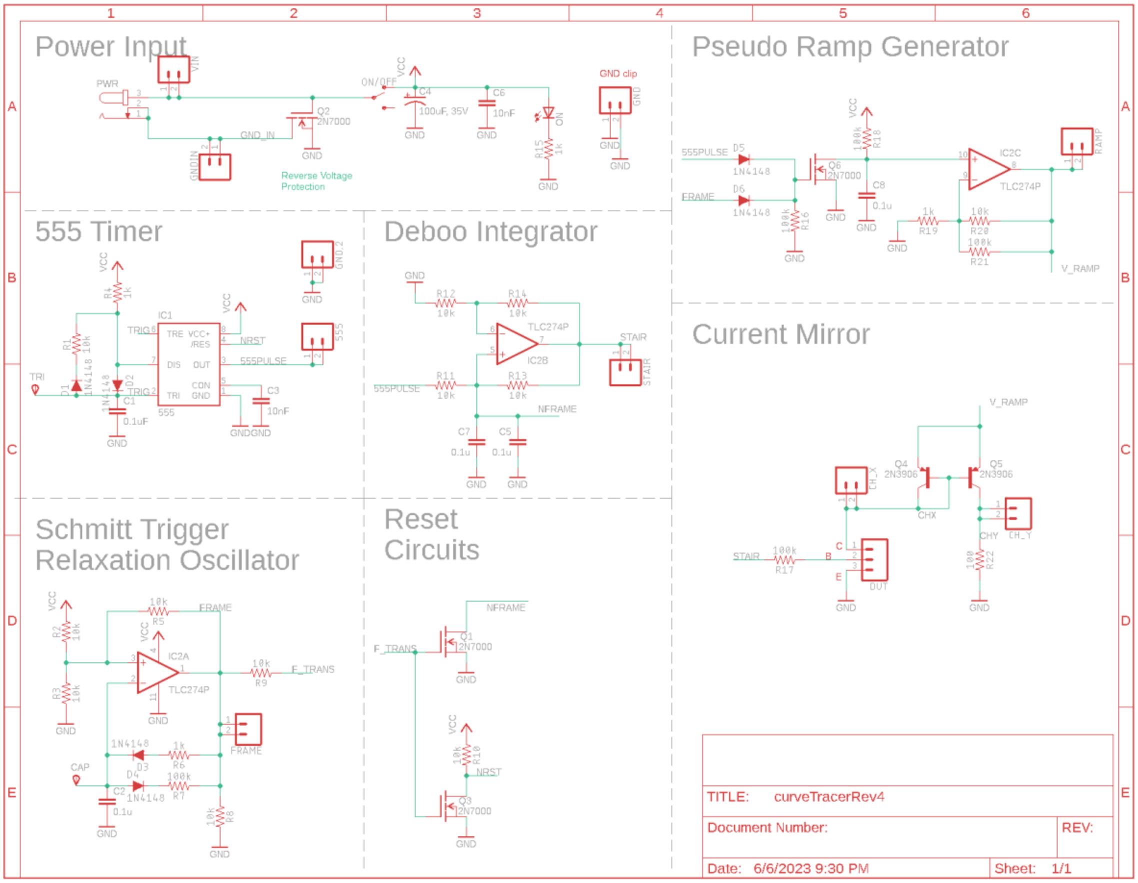


Figure 8: The schematic for the BJT curve tracer.

POWER INPUT Subsystem

The POWER INPUT area of the schematic in Figure 8 enables you to supply power to the BJT curve tracer board from either an power jack or through the VIN and GND terminals. A SPDT slide switch allows the user to turn the BJT curve tracer on or off while leaving the power input connected. An indicator LED confirms the BJT curve tracer is powered on when it is illuminated. A pair of capacitors smooths out the power delivered to the BJT curve tracer. Finally, a MOSFET disconnects the power input if you accidently reverse the positive and negative power leads. You will learn more about this MOSFET in a later lab.

555 TIMER Subsystem

The 555 TIMER subsystem in Figure 8 is identical to the circuit analyzed in in Figure 4.

Parts Designators

Most of the parts in this schematic have a designator and a value. The designator is a letter followed by a number. The designator letter tells you what type of part it is, “R” for resistor, “C” for capacitor, etc. The designators on the PCB match those on the schematic in Figure 8. This provides you with a means to find a part. You should take a moment to notice parts which are logically related in the schematic are physically proximal on the PCB.

Polarized Parts

Most of the parts you will solder into the PCB can be installed in more than one way. Parts which must be installed in a correct orientation are called polarized. If you would like to more information about the polarized parts you will install in your board, please check the Assembly Guide.

Test Points

Test points are electrical connections to important points in the circuit and frequently are the target of test and measurement equipment. All the test points on the BJT curve tracer PCB are formed with pairs of pads that you will connect together with an approximately 1 cm long piece of wire – you can use the cut off end of resistor lead for this. These wire loops make it easy to connect oscilloscope probes and alligator clips to your to your circuit. Consult the Assembly Guide for a complete list of test point connections that you should solder this week.

Part Identification

The BJT Curve Tracer has many loose thru-hole resistors of different values. It is worth your time to become familiar on how to interpret the resistors' color bands. We will use the handy table in Figure 9 as our guide.

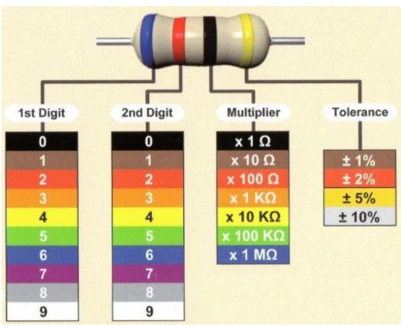


Figure 9: The colored bands on resistors indicate their resistance value.

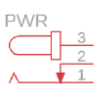
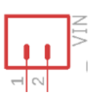
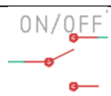
The resistors provided with the curve tracer have 4 colored bands. When looking at a resistor, hold it so that the metallic gold band is to the right – all our resistors have 5% tolerance. With the resistor correctly positioned, read the color bands from left to right, converting the colors into the numerical codes given in Figure 9. For example, let's say that from left to right the color bands are red, red, orange, gold. This pattern converts to 22x1kΩ ±5% The first three bands are the resistance, in this case 22 kΩ. The fourth band is the range of resistances you can expect the resistor to have. This phrase means the actual resistance of the example resistor could be as low as 22 kΩ – 0.05*22 kΩ = 22 kΩ – 1.1 kΩ = 20.9 kΩ or as high as 23.1 kΩ. Check your understanding by filling in the blank space in Table 2 with the correct color or resistance code.

Table 2: Complete the missing entries in the table of resistance color codes.

Value	Band 1	Band 2	Band 3	Band 4
220				
2.2k				
3.3k				Gold
4.7k		Purple		
6.8k				
10k				
15k	Brown			
33k				
47k			Orange	
100k				
470k				

To make sure you can positively identify all the elements in the schematic complete Table 3 by filling in the **Match** column with the letter corresponding to the **Schematic Symbol** for a **Physical Part**.

Table 3: Match the schematic symbol with the corresponding part.

Schematic Symbol
A 
B 
C 
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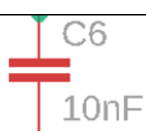
E 
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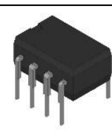
Match	**Physical Part**
	
	
	
	
	
	

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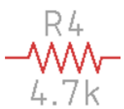
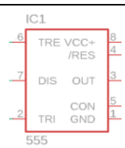
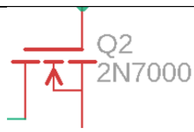
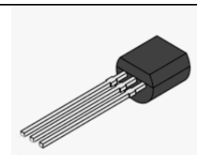
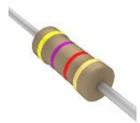
H	
I	
J	

J	
	
	
	

Testing

You should take care and align the resistors so their gold tolerance bands all facing the bottom of the board. This positioning will make it easier to read their values and compare your work to pictures of the assembled board. Look at the **How To: Solder** posted on Canvas for recommendation on how to solder through hole parts.

Once you have completed assembly of your POWER and 555 TIMER subsystems, perform the following test.

1) Power up an oscilloscope, attach a probe to Channel 1 and configure it as follows.

Ch 1 probe	555PULSE testpoint
Ch 1 ground clip	BJT ground loop
Horizontal (scale)	200 µs
Ch 1 (scale)	1V or 2V (whatever fits better)
Ch 2 probe	TRI pin of 555 Timer
Ch 2 (scale)	2V
Trigger mode	Auto
Trigger source	Ch 1
Trigger slope	↑
Trigger level	4.5V

2) Set the GND reference of Ch 1 and Ch 2 to the lowest visible reticule. The waveforms will overlap the same as they did in the LTspice simulation. The TRI pin of the 555 Timer is available by remove the end of the oscilloscope probe and putting the tip into the yellow circled pad in Figure 10. Note that this pad is labeled "TRI".

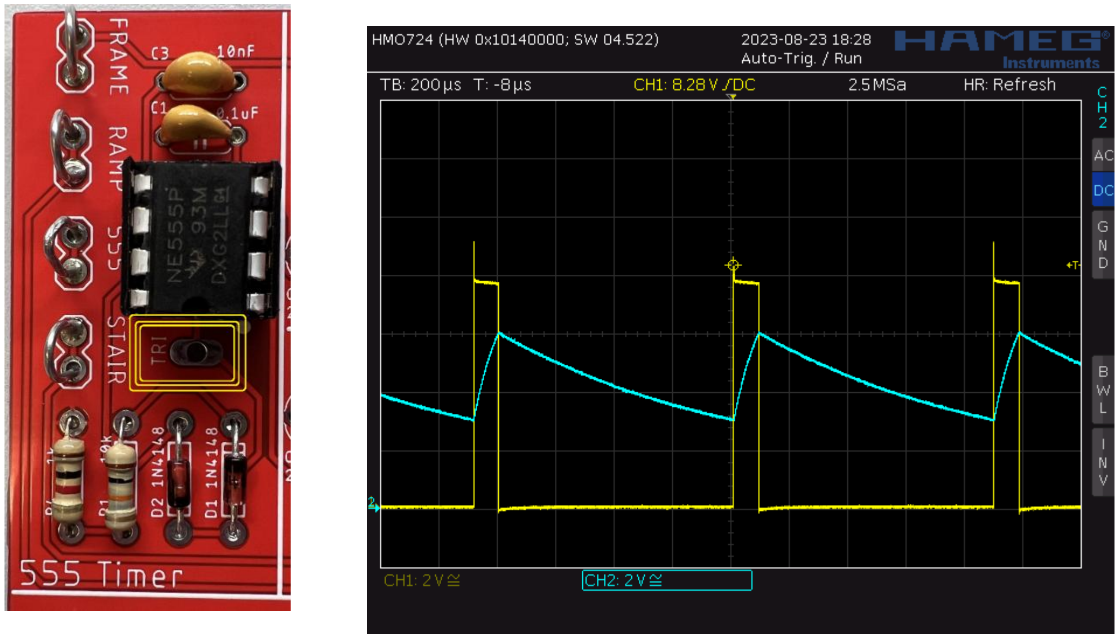


Figure 10: (Left) The TRI pin of the 555 Timer is available at the circled pad. (Right) An oscilloscope trace showing the two output you need to capture. Note that this image was captured on a Rhode&Schwarz HMO724.

3) Take a screen shot of the two waveforms and include in your lab report. Screen shot the oscilloscope traces on USB. Cell phone pictures will lose points.
[Save] → <u>Save</u> → <u>Format</u> → <u>24-bit Bit... (*.bmp)</u> [Save] → <u>Save</u> → <u>Press to Save</u>

Comparison: 555 Timer

Complete the **Analysis, Simulation** and **Empirical** columns of the following table using the information you found throughout this lab. Represent your answer to 3 significant figures using the units given in parenthesis in the **Quantity** column. You will need this table in later labs, so keep it handy.

Table 4: Comparison of the timer output from analysis, simulation and the actual circuit.

Quantity	Analysis	Simulation	Empirical
Time high (µs)			
Time low (µs)			
Period (µs)	660 µs		
Frequency (kHz)			
Duty Cycle		8.70%	

Turn In: 555 Timer

1) Make a record of your response to numbered items below and turn in a single lab report for your team on Canvas using the instructions posted there.
2) Include the names of both team members at the top of your solutions.
3) Use complete English sentences to introduce what each of the items listed below is and how it was derived.

Hint, use Ctrl+click to follow links. This also works for all the Figures and Tables in these labs.

Analysis	Question 1, 2	Voltage to assert S and R
Analysis	Question 3	SR Latch truth table
Analysis	Question 4	V_t vs. TRI state
Analysis	Question 5	Time to charge 3V to 6V
Analysis	Question 6	Time to discharge 6V to 3V
Analysis	Question 7	Behavior when V_t is less than 3V
Analysis	Question 8	Behavior when V_t is greater than 6V
Analysis	Question 9	Plot V_t vs. *time*
Analysis	Question 10	Determine rise, fall and period
Analysis	Question 11	Frequency of waveform
Analysis	Question 12	Duty cycle of the OUT pin
Simulation	Question 1	Simulation waveforms
	Question 2	Show OUT and TRIG waveforms
	Question 3	Measure OUT waveform characteristics
	Question 4	Measure TRIG waveform characteristics
Empirical	Table 2	Resistance color code
	Table 3	Parts Identification
	Screen capture	of 555 Timer output
Comparison	Table 4	Compare timer output in different models.